Conditional generation of cathode vacancy patterns: separating latent learning, spatial decoding, and relaxed energy

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Abstract

We test whether a conditional variational autoencoder (CVAE) proposes new Li/vacancy configurations at a requested composition and energy range more efficiently than uniform sampling and mutation. Earlier spatial decoding improved teacher-defined yield, but its latent variables were nearly unused. The present matched controls separate a graph-posterior CVAE, a no-latent autoregressive sampler, and a dense-posterior CVAE with the same causal spatial decoder. Nine 4,000-update fits and fresh development scoring are complete. Spatial AR matches the graph CVAE, while a dense posterior retains information without useful prior-generation gain. Relaxed-endpoint acquisition continues. Exhaustive small-cell enumeration quantifies approximate symmetry and teacher disagreement. No new DFT or interatomic-potential fine-tuning is performed, and generative, numerical, and physical validation remain distinct.

1 Motivation, revised question, and independent data

Direct recycling aims to restore useful cathode composition and structure rather than dissolve the material into its constituent elements. Hydrothermal processing and annealing have regenerated spent $LiCoO_2$ (LCO) experimentally [1]. Our longer-term motivation is to select informative atomistic configurations for reference calculations and improved recycling models. The class project studies conditional generation of vacancy patterns; it does not simulate a hydrothermal liquid or establish optimal reactor conditions.

Density functional theory (DFT) supplies an approximate electronic-structure reference at specified atoms, positions, and cell. A machine-learned interatomic potential (MLIP) predicts potential energy and its force/cell derivatives. A single-point prediction leaves geometry unchanged. Relaxation moves atoms toward a local minimum; molecular dynamics (MD) uses forces to propagate them in time. A sampled vacancy mask performs none of these operations automatically.

Li/vacancy ordering changes energy even at fixed composition [2]. The working parent is $Li_{48}Co_{48}O_{96}$. An occupation vector $o \in \{0,1\}^{48}$ has $o_j = 1$ for a present Li at parent site j and $o_j = 0$ for a vacancy. Its count is $n = \sum_j o_j$; the material contains $N_a = 144 + n$ atoms and Li fraction $x = n/48$. Primary counts are $n = 12, 24, 36$. The parent geometry and cell are fixed for deletion labels. The primary teacher is UMA-S-1.2.1/`omat`; the independent teacher check is MACE-MH-1/`omat_pbe`. Both use FP32, TF 32 off, with checkpoint and inference receipts. A missing precision field in an old condition schema is not evidence that its predictions used another precision.

The parent's 192 retained site permutations define an *orbit*: all vacancy masks related by those operations. The lexicographically smallest transformed vacancy string identifies the orbit. Grouping precedes splitting; symmetry augmentation supplies equivalent views, not additional independent examples. The fixed round-0 dataset contains 40,157 TRAIN, 4,967 VAL and 5,123 TEST orbits. Only TRAIN supplies fit examples and class thresholds; VAL selects checkpoints. TEST labels are not used for these decisions. The new development proposal history was frozen at 324,970 known orbits before its proposals were scored.

The revised scientific question is whether an improvement is due to the CVAE latent pathway, to causal spatial autoregression, or to their combination, and whether a deletion-energy advantage survives actual geometry relaxation. This is more discriminating than comparing a CVAE only against uniform masks.

2 Model, conditions, and controlled alternatives

The condition is $c = (n/48, b)$, with energy class $b \in \{\text{low}, \text{middle}, \text{high}\}$. Within each count, TRAIN energy quantiles q_{20}, q_{80} define the frozen class boundaries. Energy is in eV per present atom for class assignment; the teacher, parent, geometry convention and thresholds belong to one condition schema. A low request is a supported energy *range*, not a guarantee of a global minimum or an energy-gradient optimization.

An encoder produces a diagonal Gaussian $q_\phi(z \mid o, c) = \mathcal{N}(\mu_\phi, \operatorname{diag}\sigma_\phi^2)$, where $z \in \mathbb{R}^4$ is the latent vector. The reparameterization $z = \mu + \sigma \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, permits stochastic training. At generation, a fresh $z \sim \mathcal{N}(0, I)$ is used without a target mask. The conditional autoregressive decoder factors

$$p_\theta(o \mid z, c) = \prod_{i=1}^{48} p_\theta(o_i \mid o_{<i}, z, c, h_i). \tag{1}$$

Here $o_{<i}$ is the already assigned prefix, and h_i summarizes local occupied, vacant and unassigned sites. At training, the true prefix is supplied (teacher forcing); at generation, earlier sampled decisions form the prefix. The future target occupations are excluded from h_i. If the remaining required Li count is zero, later sites are vacant; if it equals the remaining site count, all are occupied. This enforces $\sum o_i = n$ exactly.

For one training row, the loss is

$$\mathcal{L}(o, c) = -\mathbb{E}_{q_\phi} \sum_i \log p_\theta(o_i \mid o_{<i}, z, c, h_i) + \beta_t D_{\mathrm{KL}}(q_\phi(z \mid o, c) \| \mathcal{N}(0, I)) \,. \tag{2}$$

The first term rewards mask reconstruction; the KL term regularizes the posterior toward the sampling prior. Both are measured in nats with natural logarithms. The same row weights multiply reconstruction and the sum of KL over latent components. Class balancing changes TRAIN weights only; VAL retains its base weights and fixed Monte Carlo noise. The legacy block-update KL warm-up and selection objective with $\beta = 0.1$ are matched across controls.

Control	Posterior / latent	Shared decoder and discriminating question
A	Graph encoder, $z \in \mathbb{R}^4$	Physical causal neighborhoods. Does the spatial CVAE use a latent representation?
B	No encoder or latent	Same causal neighborhoods. Is a conditional spatial AR sampler sufficient?
C	Dense encoder, $z \in \mathbb{R}^4$	Same causal neighborhoods. Does an informative dense posterior improve prior sampling?
D	No latent, matched B	Fixed nonphysical association of decoder graph kernels. Three fitted screen replicates; rescoring pending. Full kernel degree/weights are retained, but prefix exposure is also audited.

The graph encoder passes messages on the full parent, including host species and vacant mobile sites, then pools host/mobile features [3]. The causal decoder uses only prefix-derived features on that fixed graph. This is not a coordinate-generating equivariant model. The software accepts a different parent/mobile ion, but a new material or cell size has not been validated.

KL, per-component $\mathrm{Var}(\mu_j)$, resampling z with shared decoder uniforms, and changing b with shared z/uniforms test different properties. A small reconstruction loss alone does not establish useful latent codes [4]. An update-based cyclic KL screen is a bounded candidate [5], not an assumed cure for collapse or evidence of better sampling.

3 Evaluation and current fitted-model evidence

Uniform masks select exactly n sites. Mutation starts from a frozen TRAIN seed in the requested bin and swaps equal numbers of present/vacant sites. Seed information, proposal counts and swap budgets are declared. Neither baseline is credited with a new unique structure when it merely repeats one already seen. Every attempted proposal remains in the denominator, including duplicates, unsupported/failed requests and invalid outputs.

For A_{snb} attempted proposals from replicate s, count n and requested bin b, the primary score is

$$Y_{snb} = \frac{\#\{\text{distinct valid orbits, new to frozen history, teacher bin} = b\}}{A_{snb}}. \tag{3}$$

The balanced yield averages all nine count/bin cells equally and is then reported per fitted replicate. The historical low-energy score using its original round-0 q_{20} remains a separately named metric. Best energy versus attempts and versus actual new teacher queries are secondary search metrics; they need not agree with requested-bin yield. Shared queries are deduplicated physically without changing attempted-proposal denominators.

Paired condition-response tests hold z and decoder uniforms fixed while changing the requested energy class; independent MLIP rescoring determines the realized energy shift. Latent-response tests change z while fixing condition/uniforms. Current paired probes report raw Hamming differences and orbit equality, failures, novelty, energy distributions and composition-resolved results. A separate FINAL distance audit will minimize over all 192 operations. A few fit seeds give descriptive variation, not a calibrated confidence interval over all future fits.

Control	Fits / updates	VAL reconstruction	VAL KL	Active latent dims
A: graph CVAE + context	3×4000	22.633–22.663	0.006–0.019	0/4
B: spatial AR	3×4000	22.618–22.643	0	none
C: dense CVAE + context	3×4000	16.275–17.148	12.515–13.322	4/4

Selected-checkpoint ranges are validation nats/row, not prior-generation scores. Activity uses $\mathrm{Var}_{\mathrm{VAL}}(\mu_j) > 0.01$; all A components also fall below 0.001 and all C components exceed 0.1. These are descriptive diagnostic cutoffs, not physical convergence thresholds.

	Balanced novel in-bin yield(%)		Original-q_{20} LOW yield(%)	
Method	UMA	MACE	UMA	MACE
A: graph CVAE	61.92	63.48	63.72	68.40
B: spatial AR	62.50	64.00	64.06	68.75
C: dense CVAE	31.48	31.31	32.64	36.98
Mutation	56.37	56.25	50.00	50.87
Uniform	33.33	33.33	18.23	18.06

These are **development** means over three fit/sampler replicates, not fresh FINAL results. UMA balanced descriptive seed SD is 1.41/1.08/0.89/1.65 percentage points for A/B/C/mutation. Uniform reuses its draws across requested bins, so its balanced mean is exactly one third here; this does not imply zero uncertainty in its LOW yield. With paired randomness, UMA LOW→HIGH shifts A/B energies by 5.35/5.45 meV/atom, versus 0.41 for C. Changing z changes A by 0.764 sites on average (69.44% same orbit), versus 13.57 sites for C (0.52% same orbit). Active C latents therefore generate diversity without useful energy-bin control in this study.

The old assessment found a spatial graph/context model at 62.04% balanced novel requested-bin yield under UMA versus 58.62% for mutation, while its latent KL was near zero. That historical assessment motivated A–C; it is not the new comparison and cannot serve as an untouched test again. The old middle-bin result was weaker than mutation. The current controls show that A and B have similar validation reconstruction and development yield. C reconstructs better while paying substantial KL, but generates no better than uniform under the balanced metric. This supports causal spatial decoding as the main measured mechanism; it does not establish a useful latent-generation advantage.

The new development pool freezes 8,640 primary attempts and 1,152 separate latent-response probes before labels. All candidate labels are now complete under both teachers. Development outcomes may select a bounded screen; a fresh FINAL manifest, thresholds, proposal/history identities and selected checkpoints must be frozen before final labels. A new proposal is novel relative to that full history, including previously inspected assessments and new acquisitions, rather than only relative to TRAIN. MACE checks use the same frozen generated geometries and its own teacher thresholds. Teacher agreement is a robustness check, not calibrated uncertainty or DFT validation.

4 Paired deletion/relaxation target and exact small-cell check

For the same initial mask o, define $E_{\mathrm{del}}(o)$ at literal parent coordinates and $E_{\mathrm{end}}(o)$ after fixed-cell atomic relaxation. The latter is a reached local endpoint, not a certified global minimum. Two models will have identical input masks, splits, architecture, initialization policy and update budget, but distinct TRAIN-fitted condition thresholds for these two energy targets. Both generators must be evaluated under *both* energy definitions. A mask generator conditioned on endpoint energy still outputs a pristine-derived deletion geometry until relaxation is actually run.

The frozen acquisition has 6,000 fresh-to-intake uniform raw masks after orbit deduplication: per count, 1,200 TRAIN, 400 VAL and 400 TEST. The pilot contains the first 32 TRAIN masks/count. All 96 pilot relaxations converged at the historical 0.05 eV/Å maximum-force target, using 3,864 actual E/F calls. This pilot has no VAL support and therefore does not qualify for target-model training. The full acquisition is in progress; interrupted attempts are preserved. The operational fit gate requires at least 30 independent low-class masks in each count's TRAIN and VAL, with all TRAIN bins represented. This is a support heuristic, not a formal power calculation. TEST energy labels remain sealed.

The complete queue reserves endpoint assessment before optional acquisition growth. The mandatory current-method LOW comparison includes spatial CVAE, spatial AR, mutation and uniform: 64×3 counts$\times$3 replicates$\times$4 methods$= 2,304$ attempts before physical deduplication. Tightening a fixed development subset to 0.01 eV/Å probes rank/bin sensitivity. Variable-cell and fixed-cell endpoints are never pooled. MACE single points at UMA endpoints are distinct from independently relaxed MACE endpoints. Site reassignment and continuous geometry distinguish occupied-site orbits from physical PES basins.

A separate low-cost check exhaustively scores all 220/924/220 raw masks at 3/6/9 Li in a stored Hu 12-site parent, once per teacher plus repeat controls. Its retained permutations match coordinates only approximately (maximum species residual 2.64×10^{-4} Å); hence all raw geometries are scored rather than assigning one representative energy to every group member. The 2,752 new E/F/stress evaluations completed without failures. At 433.15 K:

Li count	Raw masks	Orbits	S_{UMA}/k_B	S_{MACE}/k_B
3	220	6	4.176	4.167
6	924	17	4.396	5.378
9	220	6	3.121	4.133

Here $p_i \propto \exp[-(E_i - E_{\min})/(k_B T)]$ uses total-cell energies, and $S = -k_B \sum_i p_i \log p_i$ is discrete configurational entropy in eV/K per cell. Each teacher/count is normalized separately. UMA and MACE choose different lowest mean-energy orbits at all three counts. These are exact finite static MLIP ensembles, not measured LCO entropy or 48-to-12-site network transfer.

5 Interpretation, limitations, and completion plan

The principal mechanistic distinction is between *condition control* and *latent use*. A decoder may respond well to a requested energy bin while effectively ignoring z. Conversely, an encoder can reconstruct a training mask through an informative posterior yet generate poorly from its prior. The present A/B/C development controls directly separate these possibilities. Retaining a true CVAE is useful only if its measured behavior supports the latent pathway; an AR result that matches or wins remains scientifically informative.

Pristine-derived deletions omit relaxation, local strain, magnetic degrees of freedom and competing host structures. Endpoint labels address one of these limits under a stated MLIP and fixed cell, without proving a ground state or validated chemistry. Class conditions are empirical ranges, not thermodynamic populations. The exact small-cell probabilities provide a finite-model reference for raw reference-pool error, not a justification for treating a generated proposal histogram as an equilibrium distribution.

The separate research supplement preserves a first NEB route hitting its 500-step cap and another interrupted by scheduler fail-fast, alongside a numerically converged second event. A common-setting continuation/image study is ongoing. A potential-energy path span in eV/hop is not a finite-temperature activation free energy, diffusivity or DFT-validated barrier. Synthetic velocities from the actual saved 537-mode harmonic spectrum test the thermal estimator against a known model; they do not measure an anharmonic correction or validate unrestricted hopping MD. These extensions do not substitute for class-model comparisons.

Required item	Status in this interim build
Matched A/B/C	Nine complete 4,000-update fits and two-teacher development scoring; no useful latent gain observed.
D / bounded improvement	Eight 4,000-update screen fits complete (D×3, remaining-count/shell features for A/B, lower KL ceiling for A); rescoring pending.
Paired label target	96/96 pilot converged; 6,000-mask acquisition and TRAIN/VAL support audit in progress.
Endpoint assessment	Reserved and planned; no improvement claim before actual compatible relaxations.
Fresh FINAL / MACE	Pending freeze/assessment after development selection.
Small-cell reference	Complete all-raw, two-teacher reference; finite-model scope only.

Beihao: independently rerun and interpret the matched spatial no-latent control, verify decoder causality/count enforcement, and compare condition response with latent response. Automated fits do not constitute his independent contribution. If exploring a new KL policy, keep update budgets and seeds matched and select using development evidence only.

Varad: independently replay every raw-attempt denominator, frozen-history novelty, per-seed contrast, teacher-query deduplication and paired endpoint lineage/failure account. Preserve negative middle-bin/latent/relaxation findings and distinguish proposal attempts, unique queries, actual force calls and allocated resources. Do not tune from final outcomes.

Ricardo: review the actual structures, site/host changes, force thresholds, teacher disagreement and physical scope; select later DFT checks only after the no-new-DFT restriction ends. Integrate the report and verify/manual-submit against the actual November 9,11: 59 p.m. Pacific assignment instructions. The repository supplies portable commands and evidence; no collaborator message, coursework submission, new DFT or MLIP fine-tuning is performed here.

References

[1] Yang Shi, Gen Chen, and Zheng Chen. Effective regeneration of LiCoO2 from spent lithium-ion batteries: a direct approach towards high-performance active particles. *Green Chemistry*, 20:851, 2018.

[2] Anton Van der Ven, Mehmet K. Aydinol, Gerbrand Ceder, Georg Kresse, and Jürgen Hafner. First-principles investigation of phase stability in LixCoO2. *Physical Review B*, 58:2975–2987, 1998.

[3] Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70, pages 1263–1272, 2017.

[4] Alexander Alemi, Ben Poole, Ian Fischer, Joshua Dillon, Rif A. Saurous, and Kevin Murphy. Fixing a broken ELBO. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80, pages 159–168, 2018.

[5] Hao Fu, Chunyuan Li, Xiaodong Liu, Jianfeng Gao, Asli Celikyilmaz, and Lawrence Carin. Cyclical annealing schedule: A simple approach to mitigating KL vanishing. In *Proceedings of NAACL-HLT*, pages 240–250, 2019.